

DNA Computing

From Molecular Biology to Algorithms and Computation

Canonical deep edition

Chapters 1–2

Hermon AI / coordinated research-textbook series

Development manuscript / 8 September 2026

About this development edition

This cumulative edition contains Chapters 1 and 2, not a completed book. Chapter 1's source and previously published PDF are preserved. The new chapter adds mechanisms, original vector figures, exact mathematical examples, executable references, exercises and worked solutions.

The intended standard is depth with straightforward explanation, without an artificial educational-level ceiling. Figures have editable SVG sources and semantic TXT companions. The repository also includes static state-changing keyframes, not exported moving media.

No new laboratory experiment, trained DOGMA or Hermon DNA result, GPU training run or inference-performance benchmark is reported. Computed teaching examples are explicitly distinguished from historical evidence and measured scientific results.

Review is author-agent source, mathematical, skeptical and rendered-page review, not independent specialist certification. Independent review and observation of real readers remain open. This PDF is not tagged and is not claimed to satisfy PDF/UA.

Contents

1	Computing with DNA	1
1.1	Computation in matter	1
1.2	Adleman’s experiment: the founding idea	2
1.3	The graph problem, precisely	2
1.4	From adjacency to oriented strands	4
1.5	A population is not an exhaustive search	5
1.6	Filtering: why the predicates fit together	6
1.7	Laboratory mechanisms and readable evidence	7
1.8	Parallelism and computational complexity	9
1.9	Combinatorics becomes a material budget	10
1.10	The executable logical companion	11
1.11	From one search to a field	12
1.12	Exercises	13
1.13	Worked solutions and discussion	14
2	Adleman’s experiment: from graph to molecules	15
2.1	A reconstruction, not a replacement history	15
2.2	The historical graph and its unique witness	15
2.3	Four predicates and four counterexamples	16
2.4	Sequence codes: which strands carry the edges?	17
2.5	Endpoint completion and the length invariant	18
2.6	Annealing positions ends; ligation connects them	19
2.7	A molecular population has multiplicity and missing members	20
2.8	Endpoint PCR: orientation turns recognition into enrichment	20
2.9	Size selection: seven occurrences are not seven identities	21
2.10	Two bead operations that must not be conflated	22
2.11	Sequential presence tests and their exact logical mirror	23
2.12	One computation, three linked representations	25
2.13	Graduated PCR: a position projection, not sequencing	25
2.14	Executable reconstruction and independent checks	27
2.15	Purity can improve while recovery becomes fragile	27
2.16	Material, time and the conditional factorial argument	28
2.17	What the experiment opened, and what changed later	29
2.18	Exercises	30

2.19 Worked solutions and research boundaries	31
Chapter 1 glossary	33
Chapter 2 glossary	34
Bibliography	35
Index	36

CHAPTER 1

Computing with DNA

The question

Can a collection of molecules represent possible solutions, undergo operations that remove unsuitable candidates, and leave a readable answer? We will construct the logic, connect it to molecular mechanisms, and then count the resources the construction demands.

1.1 Computation in matter

A computer needs distinguishable states, rules that change those states, and an interpretation of the result. None of these requirements says that the states must be voltages. A DNA sequence can represent an object; selective binding can recognize part of that representation; a population of strands can hold many different candidates at once. The interesting step is not writing computer symbols with four letters. It is making molecular interactions perform an operation whose result has a precise computational meaning.

Figure 1.1 separates three levels that will recur throughout this book. At the formal level, a sequence of graph vertices either satisfies a predicate or does not. At the physical level, molecules associate, dissociate, react, and produce measured signals. At the interpretive level, we decide what those signals establish. Confusing these levels makes a visually persuasive demonstration look more conclusive than it is.

FORMAL

Graph + predicates
A sequence is a witness

PHYSICAL

Strands + reactions
A sample yields a signal

INTERPRETIVE

Decode + independent check
The signal supports a claim

Figure 1.1: Three levels of a DNA computation. The formal specification defines correctness; physical mechanisms implement it imperfectly; decoding and verification connect an observation back to the specification. This is an original conceptual map, not an experimental result.

Write the chain as

$$X \xrightarrow{\text{encode}} M \xrightarrow{\text{physical process}} O \xrightarrow{\text{interpret}} \text{answer about } X.$$

Here X is an instance, M is a prepared molecular population, and O is an observation. An encoding is not an implementation of the middle arrow. Conversely, an impressive chemical transformation is not yet a computation of a specified function. The design problem is to make the whole chain reliable enough for its intended use.

This chapter enters the subject through graph search. We assume ordinary programming and discrete mathematics, but develop the molecular interpretation as it becomes necessary. The companion program models the predicates, not reaction kinetics.

1.2 Adleman's experiment: the founding idea

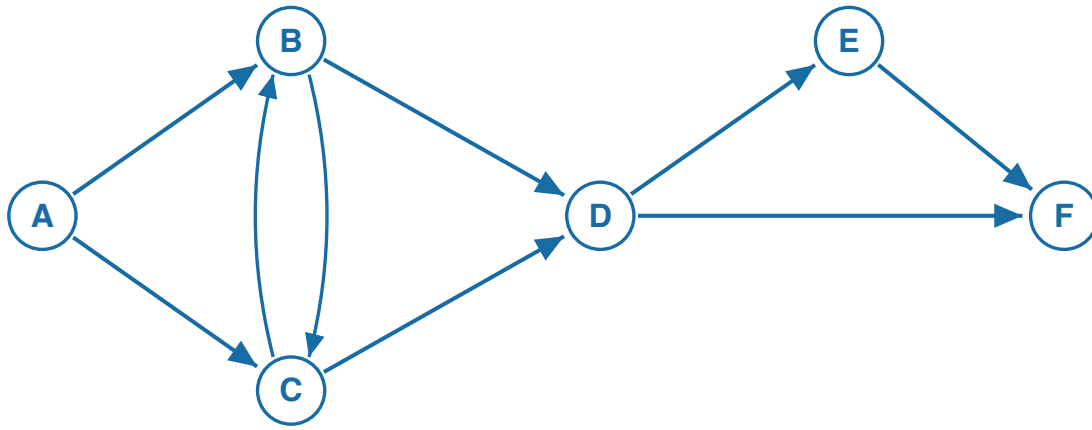
In 1994, Leonard Adleman reported a DNA implementation of a seven-vertex directed Hamiltonian-path instance. Vertex identities used 20-nucleotide codes; edge fragments and complementary vertex splints supported ligation. Endpoint PCR, a 140-base-pair size selection, and successive vertex-presence selections narrowed the material. Graduated PCR and gel patterns supported the route 0, 1, 2, 3, 4, 5, 6. This was not modern sequencing and not a general efficient algorithm for arbitrary large graphs.[1]

The experiment made a change of substrate concrete: a combinatorial search could be organized as preparation, molecular transformation, selection, and interpretation. Its significance does not depend on beating an electronic computer on seven vertices. It depends on showing that the proposed mapping can survive contact with physical operations.

Our six-vertex graph below is original teaching material. It is deliberately different from the historical instance: it has two Hamiltonian witnesses, a cycle, short routes, and a useful false candidate. We use it to derive the logic and expose assumptions. The symbolic domains in our figures are not validated laboratory oligonucleotides, and our numerical tables are program outputs rather than measurements.

1.3 The graph problem, precisely

Begin with Figure 1.2. An arrow from B to C permits that step; it does not automatically permit a step from C to B . In this particular graph both arrows exist, so a candidate may revisit that pair indefinitely unless another constraint stops it.



Start A; finish F. Nine directed edges; arrow direction matters.

Figure 1.2: The original teaching graph. Its vertices are A through F , with fixed start A and finish F . The separate arrows between B and C create a directed cycle. The graph has nine edges and exactly two Hamiltonian paths with these endpoints.

Formally, $G = (V, E)$, where

$$V = \{A, B, C, D, E, F\}, \quad E = \{AB, AC, BC, BD, CB, CD, DE, DF, EF\}.$$

The shorthand AB means the ordered pair (A, B) , not an undirected connection.

Definition 1.1. A directed **walk** $P = (v_1, \dots, v_k)$ has each $v_i \in V$ and each $(v_i, v_{i+1}) \in E$. A simple directed path is a walk with no repeated vertex. A **Hamiltonian path** visits every vertex exactly once. With fixed endpoints s, t and $n = |V|$, it satisfies

$$k = n, \quad v_1 = s, \quad v_n = t, \quad v_i \neq v_j \ (i \neq j), \quad (v_i, v_{i+1}) \in E \ (1 \leq i < n).$$

We retain the word *walk* for candidates that repeat a vertex. This matters because a molecule does not know that revisiting B is forbidden merely because our problem uses the word path.

For $ABCDEF$, each adjacent pair is legal and all six vertices occur once. For $ABCBDF$, all adjacent pairs are still legal, the endpoints are right, and the length is six, but B occurs twice and E is absent. For $ABDEF$, the edges and endpoints are legal but C is missing. For $ABECDF$, all vertices occur once, yet BE and EC are not edges.

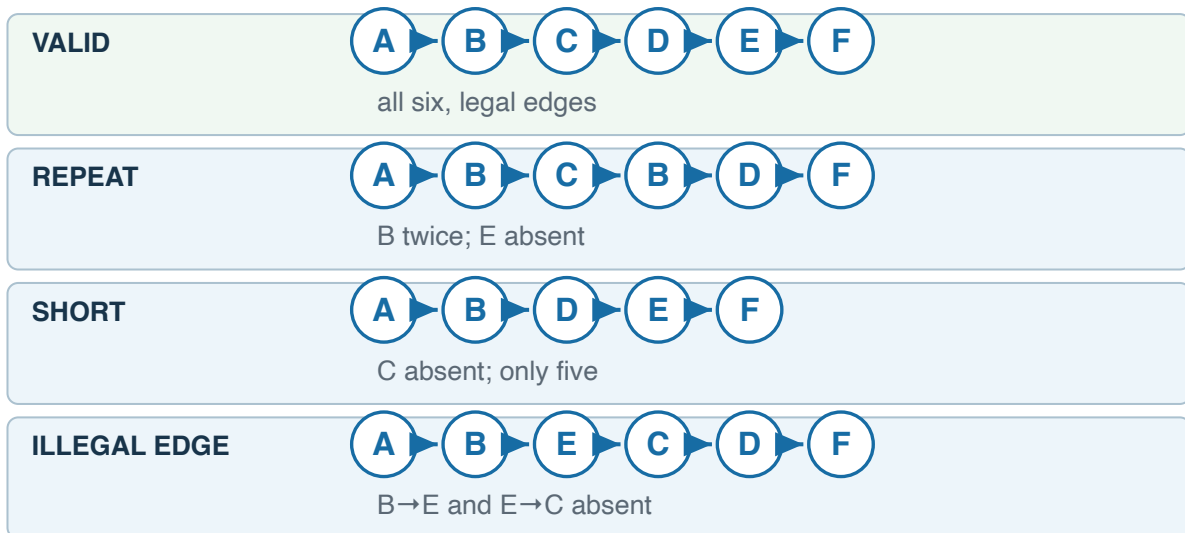


Figure 1.3: Four candidates isolate four logical issues: a witness, a repeated vertex, insufficient length, and illegal adjacency. A presence test cannot detect an illegal edge merely by checking that every vertex occurs.

The **decision problem** asks whether a witness exists. The search problem asks for one. Once a candidate order is available, its verification is straightforward; finding that order without being given it is the difficult part. We must not transfer the ease of verification to the cost of producing a witness.

1.4 From adjacency to oriented strands

DNA's sugar–phosphate backbone has a direction, conventionally written 5' to 3'. Complementary strands align antiparallel, with A paired to T and C paired to G.[2] In our diagrams, letters such as b_L denote whole sequence domains, not individual bases. A domain is a contiguous sequence chosen to play a particular recognition role.

Split a vertex code into two equal-length domains:

$$O_v = L_v R_v, \quad |L_v| = |R_v| = h.$$

An internal edge $u \rightarrow v$ is represented by the 5'–3' fragment $R_u L_v$. The next edge $v \rightarrow w$ uses $R_v L_w$. A complementary copy of the vertex code $L_v R_v$ can align the end of the first fragment with the beginning of the second. Figure 1.4 shows this meeting at B .

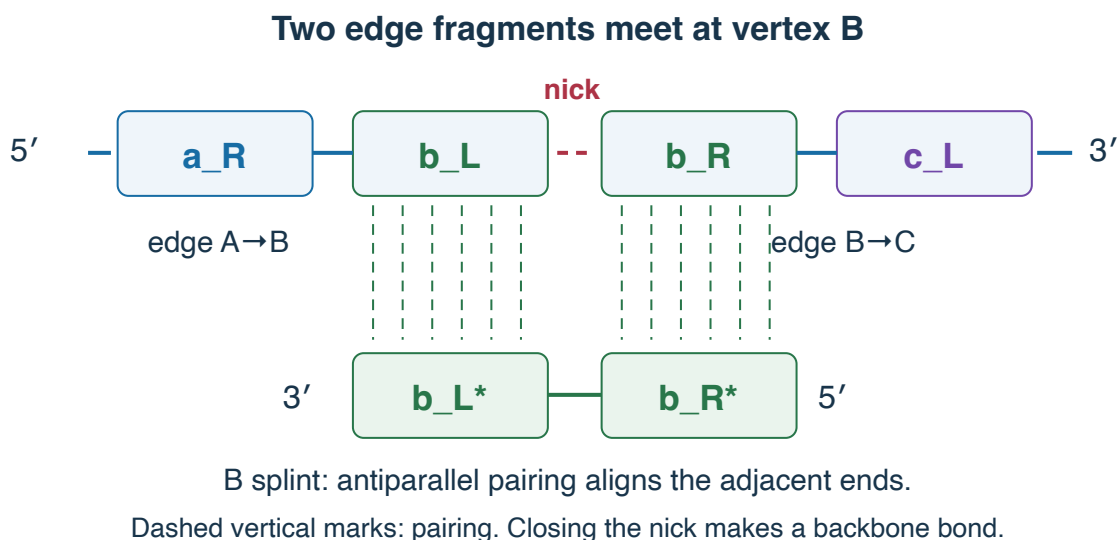


Figure 1.4: Domain-level edge encoding at vertex B . The top fragments represent $A \rightarrow B$ and $B \rightarrow C$. The bottom complementary vertex splint runs antiparallel. Pairing positions the adjacent ends; ligation closes their nick. Domain widths are schematic, and chemical end groups are omitted.

This orientation prevents a common explanatory reversal. In this construction, complementary *vertex* strands align *edge* fragments. It is not a picture of arbitrary whole-vertex strands being glued together by a vague edge label.

A domain marked with a star is shown in its paired, antiparallel orientation. If all sequences are instead listed 5′–3′, the partner must be written as a **reverse complement**. For example, the reverse complement of 5′-ACGA-3′ is 5′-TCGT-3′, not AGCT and not TGCT.

Endpoint fragments need a convention too. Complete the first fragment with all of O_s and the last with all of O_t , rather than retaining only their interior-facing half-domains. For a route with k vertices, the resulting idealized single-strand code is $O_{v_1} \cdots O_{v_k}$, of length $2hk$ nucleotides. Length then becomes a proxy for the number of vertex occurrences. Without the endpoint convention, the length formula changes.

This construction explains the intended correspondence, not sequence-design success. Unwanted partial matches, folded structures, and different binding strengths can break the clean symbolic picture. Later chapters on molecular representation, hybridization, thermodynamics, and sequence design will replace the domain abstraction with physical constraints.

1.5 A population is not an exhaustive search

The candidate pool is best understood as a **multiset**: a route may have many molecular copies, another may have one, and a valid route may be absent. A million copies of one wrong route do not cover a million distinct possibilities.

ABCDEF	①	①	①		x3
ACBDEF	②	②			x2
ABCBDF	③	③			x2
ABDEF	④	④	④	④	x4
ABECDF	⑤				x1

Illustrative copies only; the last route is an injected pseudo-path.

Figure 1.5: An illustrative candidate population. Copy count, distinct-route count, and coverage of valid solutions are different quantities. The last route is an intentionally injected pseudo-path for testing the verification boundary; the pictured multiplicities are not experimental abundances.

Let $N(P)$ be the number of copies of candidate P . The total population is $\sum_P N(P)$, whereas its support is $\{P : N(P) > 0\}$. Selection can enrich a member of that support. In a pure selection model it cannot recover a witness that was never present.

Our executable model generates every legal walk containing between one and seven vertices, starting from any vertex. The bound is explicit because the B, C cycle allows arbitrarily long walks. There are 106 such bounded walks. This finite enumeration is exhaustive within its declared domain; a prepared physical population is a sample and should not inherit that guarantee.

We will also inject one copy of $ABECDF$. This sequence is not generated by the ideal edge-valid walk generator. It stands for a representation error that local presence and length tests alone cannot exclude. Its role is logical fault injection, not a quantitative model of contamination.

1.6 Filtering: why the predicates fit together

The intended logic is easy to state after the encoding is fixed. For a candidate $P = (v_1, \dots, v_k)$ define

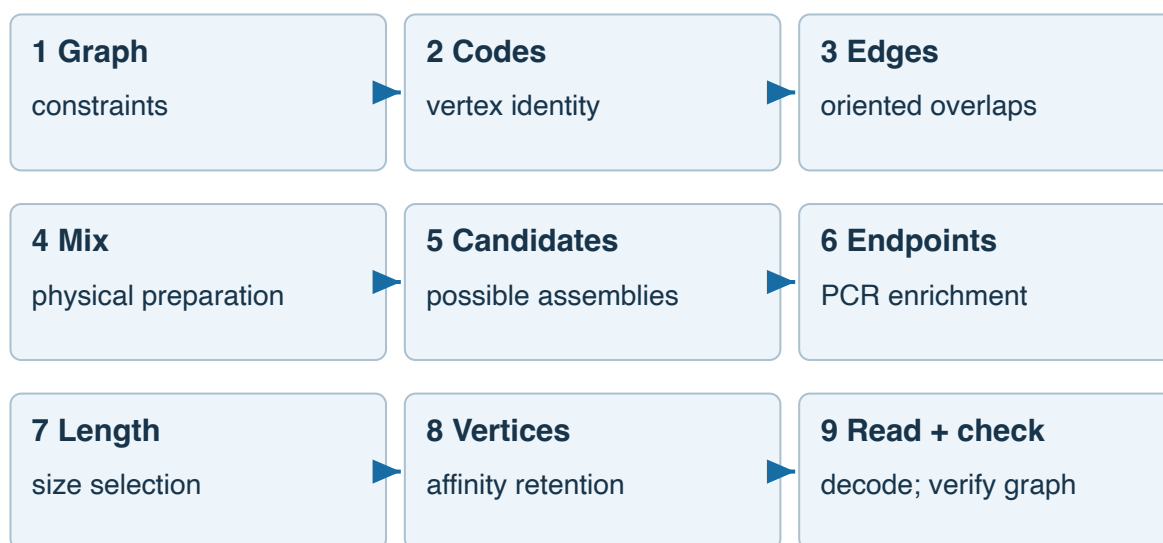
$$\begin{aligned}\mathcal{E}(P) &= [v_1 = A \wedge v_k = F], \\ \mathcal{L}(P) &= [k = 6], \\ \mathcal{C}(P) &= [\text{every vertex in } V \text{ occurs in } P].\end{aligned}$$

Brackets denote a Boolean predicate, not a concentration. An ideal filter retains exactly the candidates for which its predicate is true.

Proposition 1.1. *If a sequence has exactly n positions and contains every member of a set of n vertices, each vertex occurs exactly once and no other symbol occurs.*

Proof. Each of the n required distinct vertices occupies at least one position. These already require all n positions. A repetition or an additional symbol would require another position or displace a required vertex. $\square$

The proposition explains why length and coverage together remove repeated vertices. It says nothing about adjacency. A final edge check is still required unless edge validity is guaranteed by the physical representation.



Design → chemistry → evidence. Panels are conceptual stages, not a lab protocol.

Figure 1.6: Nine conceptual stages from graph specification to a checked witness. The row-major order is the reading order; panel arrows connect adjacent stages within a row. These stages organize the explanation, not nine literal laboratory operations. The companion directory contains nine static focus frames.

The program applies endpoint, length, and coverage predicates, then independently verifies the decoded route against G . Its exact output is:

Stage	Legal walks	With pseudo-path
Generated	106	107
Endpoints	14	15
Length	4	5
Coverage	2	3
Verified	2	2

Both populations finish with $ABCDEF$ and $ACBDEF$. The pseudo-path survives the first three filters and is rejected only by full verification. The table makes a useful distinction visible: *passing selection is not the same event as satisfying the specification*.

Physical selection is not Boolean. If a valid molecule survives stage i with probability q_i conditional on surviving the preceding stages, its overall survival probability is $\prod_i q_i$ by the probability chain rule. Independence is unnecessary for these conditional probabilities. If one instead measures isolated-stage yields and multiplies them, transferability or independence assumptions are needed. Six hypothetical stages each retaining 90 percent leave about 53.1 percent overall. This is a sensitivity calculation, not an estimate of Adleman's yield.

1.7 Laboratory mechanisms and readable evidence

Hybridization is complementary association. **Ligation** makes a covalent backbone connection between suitable adjacent ends; T4 DNA ligase can seal a nick between a 3' hydroxyl and a 5'

phosphate in a duplex.[3] Pairing and ligation therefore do different jobs: one aligns, the other connects. A diagram with only base-pair contacts has not yet shown a continuous product strand.

Polymerase chain reaction (PCR) repeatedly separates strands and uses primers and polymerase to make copies of a selected region.[4] Primers complementary to endpoint regions make endpoint-compatible candidates preferentially amplifiable. This is enrichment through chemistry, not a perfect software predicate. Amplification of an undesired template can also produce a strong signal.

Gel electrophoresis separates charged molecules in a gel under an electric field; fragment size is a major determinant of migration under appropriate conditions.[5] Selecting a size fraction implements a length constraint only as accurately as the separation and recovery permit. In an affinity step, a complementary probe recognizes a required subsequence and supports retention of molecules carrying it. Recognition does not reveal the complete order of the strand.

Join	Ligase seals a nick	incompatible / unjoined ends
Endpoints	Primers + amplification	off-target amplification
Length	Gel size fraction	overlapping bands
Contains B	Complementary capture	loss / nonspecific retention
Witness	Decoded order + verifier	ambiguous mixed signal

Predicate / intended physical mechanism / possible failure

Figure 1.7: The intended operation, its physical mechanism, and a representative failure mode. The mapping is conceptual: measured specificity, yield, and readout sensitivity are needed before treating a physical step as its ideal predicate.

For readout, consider a primer anchored at the start code and a second primer associated with an internal vertex. In our idealized six-vertex, 20-bases-per-vertex model, the corresponding lengths locate that vertex in a pure witness. Figure 1.8 draws the expected lengths for *ABCDEF*.

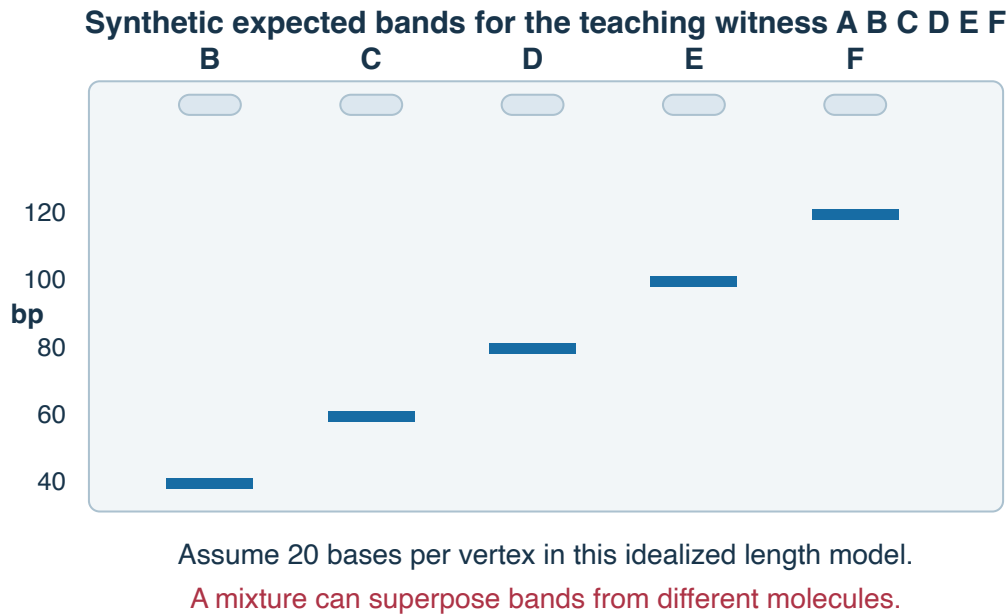


Figure 1.8: Synthetic graduated-PCR-style length readout for the original teaching witness. Lanes B through F indicate expected products of 40, 60, 80, 100, and 120 bases under the stated idealized encoding. These bands are neither photographed data nor a reconstruction of the historical seven-vertex gel.

Now mix $ABCDEF$ with $ACBDEF$. Both B and C can produce lengths associated with positions two and three. The union of bands records possibilities across molecules; it does not assign every band to one molecule. Recovering a complete witness requires resolving that ambiguity, for example through further isolation and a suitable readout, followed by checking the decoded order.

The historical report discusses imperfect separation and spurious paths as practical concerns.[1] The general inference is important: a positive, independently checked witness can establish existence; absence of a detectable signal does not establish nonexistence without a model of generation, retention, and detection.

1.8 Parallelism and computational complexity

A candidate Hamiltonian route is a short certificate. Given an adjacency lookup and a vertex-indexing structure, a verifier scans the n entries, rejects duplicates or unknown vertices, checks the endpoints, and checks $n - 1$ edges. This takes $O(n)$ lookup operations. Constructing an adjacency matrix may itself cost $O(n^2)$ space; the representation and setup belong in the resource account.

The class P contains decision problems solvable in polynomial time on a standard deterministic machine. NP contains those for which a yes-instance has a polynomial-size certificate checkable in polynomial time. NP-hardness means that all problems in NP reduce to the problem by an appropriate polynomial-time reduction; NP-completeness combines NP-hardness with membership in NP. Directed Hamiltonian path is NP-complete.[6] These definitions do not say that every instance is hard or that P has been proved unequal to NP.

A tube can contain many candidates undergoing the same operation. This changes the organization of work: material is used to hold simultaneous possibilities. But a small number of filtering stages is not a bound on the amount of material needed to make the right candidate

likely to exist. Complexity accounting must include population size, representation length, preparation, physical time, and readout.

1.9 Combinatorics becomes a material budget

Suppose the endpoints are fixed in a complete directed graph. The $n - 2$ interior vertices can occur in

$$N = (n - 2)!$$

orders. This is a count of candidate orders, not a statement that a sparse graph produces each order. Our nine-edge graph removes many orders through adjacency; the complete-graph calculation isolates how quickly a naive permutation space grows.

Vertices n	Internal orders $(n - 2)!$
6	24
10	40,320
15	6,227,020,800
20	6,402,373,705,728,000
25	25,852,016,738,884,976,640,000

Assume independent draws from a candidate-generation distribution. Let p be the probability that one draw produces a chosen target event, such as one particular witness or any valid witness. After M draws, the miss probability is $(1 - p)^M$. Requiring a miss probability at most δ , where $0 < \delta < 1$, gives

$$(1 - p)^M \leq \delta, \quad M \geq \frac{\ln \delta}{\ln(1 - p)} \quad (0 < p < 1).$$

The denominator is negative, which reverses the inequality when dividing. The minimum integer is the ceiling. When $p = 1$, one draw suffices; when $p = 0$, no finite number suffices.

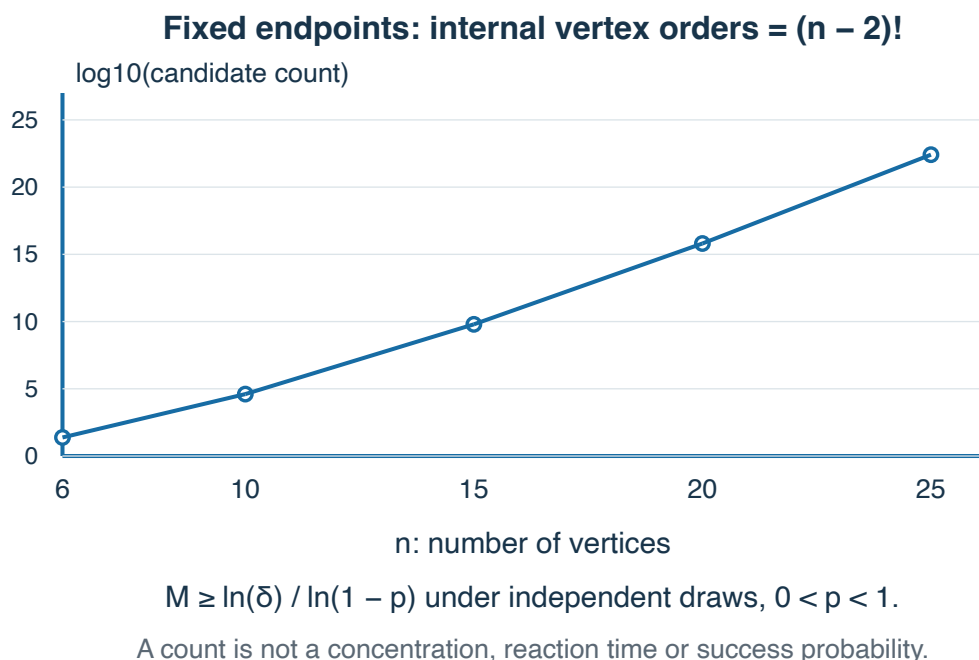


Figure 1.9: Original factorial counts plotted on a logarithmic vertical scale, together with the independent-sampling bound. Connecting segments guide the eye; no measured chemical scaling law is fitted. Neither curve nor equation assigns a concentration, reaction time, or experimental yield.

For a uniform draw from the 24 interior orders of six vertices, one designated order has $p = 1/24$. A 99-percent hit probability needs 109 draws. Our graph has two witnesses, so the target event *either witness* has $p = 2/24$ in that same hypothetical uniform space and needs 53 draws. These are different questions. The actual legal-walk generator is neither uniform over permutations nor a physical random sampler.

For small p , $\ln(1 - p) \approx -p$, giving $M \approx \ln(1/\delta)/p$. If each generated target survives independently with probability q , the effective per-draw success probability is pq . This additional model must be declared rather than smuggled into a copy count.

Molecule count connects to a physical budget through

$$M = cV_{\ell}N_A,$$

where c is amount concentration in moles per litre, V_{ℓ} is volume in litres, and $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$ is the exact Avogadro constant.[7] The units cancel to a count. Raising M by increasing concentration or volume does not guarantee unchanged reaction behavior.

A useful resource description is therefore a vector, not one speed number:

$$R = (\text{sequence inventory, molecule count, volume, time, energy, readout cost}).$$

Energy comparisons must specify whether synthesis, temperature control, separation, and measurement are included. Parallelism relocates costs; it does not erase the accounting boundary.

1.10 The executable logical companion

The implementation in `examples/deep/ch01.py` intentionally avoids an optimized solver. A bounded walk generator exposes repeated vertices, staged filtering exposes which constraints

do the work, and an independent permutation oracle checks the final answer. The excerpt below is generated from the actual source, so its line numbers and operations cannot silently diverge from the tested program.

```

15 def is_hamiltonian(route, vertices=VERTICES, edges=EDGES, start=START, end=END):
16     return (len(route)==len(vertices) and bool(route)
17             and route[0]==start and route[-1]==end
18             and set(route)==set(vertices) and len(set(route))==len(vertices)
19             and edge_valid(route,edges))

21 def walks(max_vertices=7, vertices=VERTICES, edges=EDGES):
22     """Enumerate all legal directed walks of lengths 1..max_vertices, with repeats."""
23     if type(max_vertices) is not int or max_vertices < 1:
24         raise ValueError("positive maximum length required")
25     frontier=[(v,) for v in vertices]
26     result=[]
27     for _ in range(max_vertices):
28         result.extend(frontier)
29         frontier=[r+(v,) for r in frontier for v in vertices if (r[-1],v) in edges]
30     return result

32 def filter_pool(pool, vertices=VERTICES, edges=EDGES, start=START, end=END):
33     """Do not assume generation was sound: final edge verification is independent."""
34     stages=[("generated",list(pool))]
35     predicates=[
36         ("endpoints",lambda r: bool(r) and r[0]==start and r[-1]==end),
37         ("length",lambda r: len(r)==len(vertices)),
38         ("coverage",lambda r: set(vertices)<=set(r)),
39         ("verified",lambda r: is_hamiltonian(r,vertices,edges,start,end)),
40     ]
41     for name,predicate in predicates:
42         stages.append((name,[r for r in stages[-1][1] if predicate(r)]))
43     return stages

```

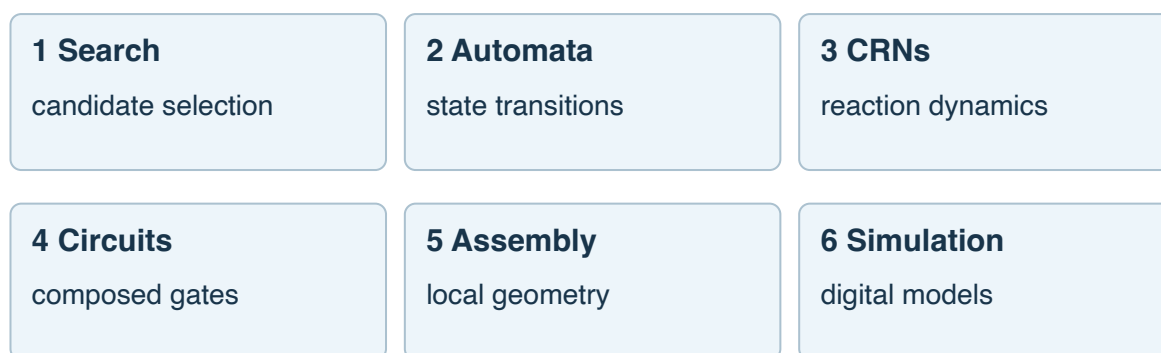
There are two independent routes to the expected survivors. The stage model starts with all bounded legal walks. The oracle fixes the endpoints, permutes the four interior vertices, and checks adjacency. Agreement on this graph catches omissions in either construction more effectively than hard-coding one expected route.

Other tests target boundaries: a repeated vertex can pass length, a pseudo-path can pass coverage, an empty route cannot be a witness, and duplicated molecular candidates should remain duplicated until an explicitly set-valued result is requested. Converting a route itself to a set too early destroys precisely the order and multiplicity the problem asks us to reason about.

The sampling helper checks its domain and is tested at the integer threshold: M must meet the requested miss bound while $M - 1$ must fail it. None of these tests establishes hybridization specificity or reaction yield. They establish consistency of an exact, inspectable logical model that a later physical implementation must refine.

1.11 From one search to a field

DNA computation is not restricted to generate-and-filter search. Benenson and colleagues demonstrated a biomolecular finite-state-machine construction.[8] Soloveichik, Seelig, and Winfree developed a DNA realization framework for chemical reaction networks.[9] Qian and Winfree demonstrated larger strand-displacement logic circuits.[10] Winfree and colleagues demonstrated designed two-dimensional DNA crystal self-assembly.[11]



Storage and learned DNA representations are related, distinct activities.

Figure 1.10: A mechanism map, not a performance ranking. Search emphasizes population selection; automata emphasize state transitions; reaction networks emphasize dynamics; circuits emphasize signal composition; assembly emphasizes spatial organization. Each needs its own representation and correctness argument.

The common idea is programmable physical organization. The hard questions then become specific. Can local recognition support reliable composition? Which errors accumulate with depth or population size? How do we restore signals, supply free energy, isolate unwanted interactions, and obtain an answer without spending more effort on readout than the computation saves?

Adleman's example gives us a disciplined starting vocabulary: specification, encoding, population, transformation, selection, observation, and verification. Chapter 2 will reconstruct the founding experiment more closely and examine what its controls and observations support. The later molecular chapters will supply the thermodynamics and kinetics deliberately abstracted here. The corresponding Evolutor book asks a different question about genomic organization in digital systems; it must not treat this physical demonstration as evidence that a software architecture works.

1.12 Exercises

1. Verify $ACBDEF$ against every condition in the definition. List the five directed edges used.
2. Prove that length plus coverage excludes repetition. Give a six-symbol counterexample showing that it does not enforce adjacency.
3. Draw the domain-level junction at C for $B \rightarrow C \rightarrow D$. Mark both strand directions. Compute the reverse complement of $ACGA$.
4. Identify the first ideal filter that removes $ABDEF$ and the first that removes $ABCBDF$, assuming endpoint filtering comes first.
5. Add ten copies of $ABECDF$ to the 106 legal walks. Predict every stage count without running the program; then verify.
6. Under uniform sampling of all fixed-endpoint interior orders, derive the draw count for a 99-percent chance of obtaining either witness.
7. Five stages each retain a target with conditional probability 0.8. Calculate overall retention. Explain why isolated-stage measurements might not justify the same product.

8. Design separate logical controls for candidate generation, selection loss, and readout failure. State what each can and cannot distinguish.
9. Compare an empty result from exhaustive symbolic enumeration with an empty physical readout. Which additional assumptions would justify the same conclusion?
10. Propose a fair resource comparison between a molecular implementation and a digital solver. Specify the instance family, success criterion, and accounting boundary.

1.13 Worked solutions and discussion

1. The edges are AC, CB, BD, DE, EF . All belong to E ; the six vertices are distinct, with endpoints A, F . Thus it is the second witness.
2. The n required identities occupy all n slots, leaving no room for a duplicate. But $ABECDF$ has length six and complete coverage while $BE, EC \notin E$.
3. The upper fragments are b_{RC_L} and c_{Rd_L} , both written $5'-3'$. A complementary c_{Lc_R} splint runs $3'-5'$ underneath and aligns the nick between c_L and c_R . ACGA has reverse complement TCGT.
4. Both pass endpoints. $ABDEF$ fails length. $ABCBDF$ passes length but fails coverage because E is absent.
5. Counts are 116, 24, 14, 12, 2 at generation, endpoints, length, coverage, and verification. All ten injected copies persist until the edge check.
6. $p = 2/24 = 1/12$; $\lceil \ln(0.01)/\ln(11/12) \rceil = 53$. This is a sampling-model calculation, not a prediction of molecular yield.
7. Retention is $0.8^5 = 0.32768$. The conditional formulation already accounts for the selected population reaching each stage. Isolated yields may differ after previous processing changes that population.
8. A known distinctive target introduced before generation tests a different chain from one introduced immediately before readout. Controls entering between selection stages can help localize loss. Their behavior need not match every unknown candidate, so they bound an inference rather than prove universal recovery.
9. Complete symbolic enumeration with a correct verifier establishes no witness in its finite search domain. A physical negative additionally needs adequate coverage, recovery, and detection sensitivity, with an explicit failure bound.
10. Use identical mathematical instances and success requirements; include preparation, failed runs, decoding, and verification. Report distributions of total cost and success, not only the fastest central operation. No winner follows from the proposal alone.

CHAPTER 2

Adleman's experiment: from graph to molecules

The problem behind the picture

A route is an ordered mathematical object. A tube contains physical strands, not routes as such. This chapter reconstructs the mapping closely enough to ask, at every stage: what property is represented, what operation acts on it, and what would make the resulting inference wrong?

2.1 A reconstruction, not a replacement history

Chapter 1 used an original six-vertex teaching graph. We now change instances deliberately: the vertices in this chapter are the seven numbered vertices of Adleman's 1994 experiment. The point is to carry the correspondence all the way from directed edges to oriented sequence fragments, laboratory selection, and readable evidence.

The historical anchor is the original report, including its experimental notes, rather than a simplified diagram of a generic DNA computer.[1] Our redrawn graph and the three published sequence codes preserve factual information from that report. Our route inventory, fault injection, mathematical proofs, sensitivity model and synthetic band diagrams are new teaching constructions. None is a measurement of the original reaction.

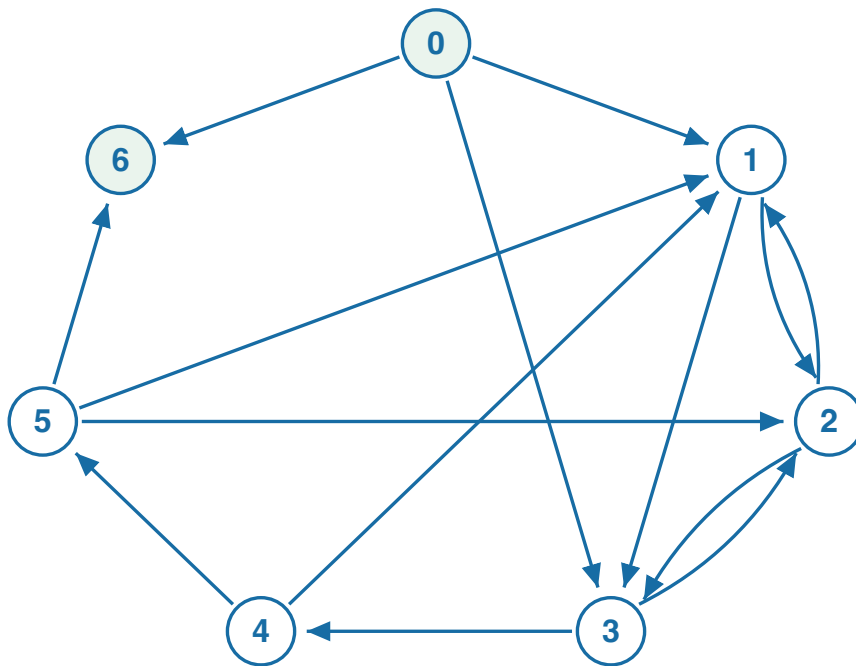
Three distinctions will organize the reconstruction. First, hybridization associates complementary strands, while ligation creates a backbone bond. Second, PCR and affinity selection enrich physical populations rather than evaluate perfect Boolean predicates. Third, a readable signal is evidence to interpret, not a substitute for checking a decoded witness. These distinctions prevent a short abstract algorithm from concealing the actual work.

2.2 The historical graph and its unique witness

Let $G = (V, E)$ with $V = \{0, 1, 2, 3, 4, 5, 6\}$, start $s = 0$, and target $t = 6$. The adjacency list below is an exact transcription of the directed graph in the original Fig. 1:

0	1, 3, 6	1	2, 3
2	1, 3	3	2, 4
4	1, 5	5	1, 2, 6
6	$\emptyset$		

Each row lists outgoing neighbors. There are fourteen edges. In particular, both $1 \rightarrow 2$ and $2 \rightarrow 1$ exist, as do $2 \rightarrow 3$ and $3 \rightarrow 2$. No reverse arrow should be inferred merely because two vertices are connected visually.



14 directed edges; fixed endpoints 0 and 6. Geometry carries no extra meaning.

Figure 2.1: Original redrawing of the seven-vertex historical instance. All fourteen directed edges are included; start and finish are labeled 0 and 6. The circular placement differs from the source layout and has no computational meaning.

The sequence $P = (0, 1, 2, 3, 4, 5, 6)$ is a witness. It is also the only witness with these endpoints. Here is a structural argument, independent of enumeration. Vertex 4 can first be reached only from 3, and vertex 5 only from 4. A Hamiltonian route cannot enter 6 early because 6 has no outgoing edge. On reaching 5 after visiting 3 and 4, taking $5 \rightarrow 1$ or $5 \rightarrow 2$ cannot lead onward to the still-unvisited finish without revisiting 3, 4 or 5. Thus the final block must be 3, 4, 5, 6. Vertices 1 and 2 must precede that block. From 0, the only way to visit both before 3 is 0, 1, 2, 3. The program separately checks all $5! = 120$ interior permutations.

2.3 Four predicates and four counterexamples

For $n = |V|$, a fixed-endpoint Hamiltonian witness $P = (v_1, \dots, v_n)$ satisfies

$$v_1 = s, \quad v_n = t, \quad (v_i, v_{i+1}) \in E \ (i < n), \quad v_i \neq v_j \ (i \neq j). \quad (2.1)$$

A candidate assembled from allowed edges may be a walk with repeated vertices. The molecule has no global instruction forbidding a revisit to 3.

Route	Edges	Occurrences	Coverage
0 1 2 3 4 5 6	legal	7	all; witness
0 1 3 4 5 6	legal	6	2 absent
0 3 2 3 4 5 6	legal	7	1 absent; 3 twice
0 1 2 4 3 5 6	illegal	7	all; not a witness

Figure 2.2: The witness, a legal short route, a legal repeated-vertex walk, and an illegal-edge permutation separate the predicates. The last sequence has every identity and the correct length, but contains forbidden transitions.

The legal short route 0, 1, 3, 4, 5, 6 omits 2. The legal seven-occurrence walk 0, 3, 2, 3, 4, 5, 6 repeats 3 and omits 1. The permutation 0, 1, 2, 4, 3, 5, 6 has all vertices, but $2 \rightarrow 4$, $4 \rightarrow 3$, and $3 \rightarrow 5$ are not graph edges. These are not three ways of saying the same thing. They show why length, coverage and adjacency need separate justification.

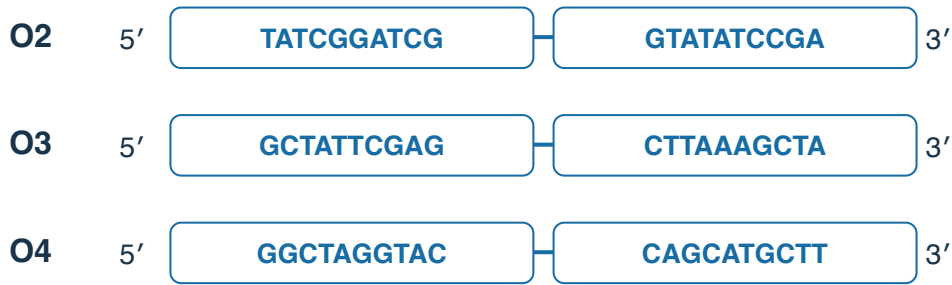
Define endpoint, occurrence-count and coverage predicates

$$\mathcal{E}(P) = [v_1 = 0 \wedge v_k = 6], \quad \mathcal{L}(P) = [k = 7], \quad \mathcal{C}(P) = \bigwedge_{v \in V} [v \in P].$$

By the counting proof in Chapter 1, $\mathcal{L} \wedge \mathcal{C}$ gives exactly one occurrence of every vertex. Adjacency remains an independent premise. In a perfect encoding it is supplied during generation; in an imperfect physical implementation it must also be checked after decoding.

2.4 Sequence codes: which strands carry the edges?

Write a vertex code as $O_i = L_i R_i$, with two ten-base domains. All codes in Figure 2.3 are written 5' to 3'. Only O_2, O_3, O_4 are shown, because these are the three vertex codes printed in the original sequence illustration. Missing historical codes are not filled in with invented sequences.



O23 = R2 L3

GTATATCCGAGCTATTCGAG

O34 = R3 L4

CTTAAAGCTAGGCTAGGTAC

Aligned splint O3*:

3' CGATAAGCTCGAATTTTCGAT 5'

Only the three printed historical codes are shown; all upper codes read 5'→3'.

Figure 2.3: Sequence-level reconstruction using the three published codes. The internal edge fragments are $O_{23} = R_2L_3$ and $O_{34} = R_3L_4$. The complementary O_3 splint is printed in aligned 3'–5' orientation. Its left-to-right letters are therefore not a 5'–3' reverse-complement listing.

The central construction is

$$O_{ij} = R_iL_j, \quad O_{jk} = R_jL_k.$$

The end L_j of the first fragment and the beginning R_j of the second together match the complementary vertex splint O_j^* . That splint brings the appropriate ends together. Replacing this with whole vertex strands connected by complementary edge bridges would describe a different construction.

For the printed example,

$$O_{23} = \text{GTATATCCGA GCTATTCGAG},$$

$$O_{34} = \text{CTTAAAGCTA GGCTAGGTAC}.$$

The twenty-base junction region across the nick is L_3R_3 , exactly O_3 . Its aligned complement is

$$3'\text{-CGATAAGCTCGAATTTTCGAT-}5'.$$

If the partner were instead listed in the conventional 5'–3' direction, its letters would have to be reversed. Direction labels are part of the information, not optional decoration.

2.5 Endpoint completion and the length invariant

Internal edges carry only code halves. The start and finish require special treatment: an edge leaving 0 carries the whole start code, and an edge entering 6 carries the whole finish code. Thus a first edge has form O_0L_j , a last edge R_iO_6 , and the direct edge $0 \rightarrow 6$ has form O_0O_6 .

Consider a route $0, v_2, \dots, v_{k-1}, 6$. After the compatible junctions are ligated, the intended upper strand is

$$O_0O_{v_2} \cdots O_{v_{k-1}}O_6.$$

There are k complete codes, so its length is $20k$ nucleotides. After complementary copying, the corresponding double-stranded product has $20k$ base pairs. For seven occurrences this is 140 bp. The units distinguish a single-strand inventory from a duplex size fraction.

This derivation depends on complete endpoints and full-length products. With only half-codes at both ends, the expression would instead be $20(k - 1)$. With truncations, nonspecific ligation or unexpected internal priming, even a correct arithmetic formula need not describe the observed fragment. A length filter inherits the assumptions of the encoding that makes length meaningful.

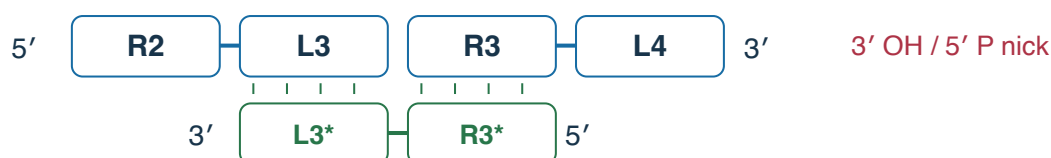
2.6 Annealing positions ends; ligation connects them

DNA strands have directional sugar–phosphate backbones. Complementary sequences can align antiparallel through base pairing.[2] The schematic domains L_3 and L_3^* stand for ten-base regions, not indivisible magnetic labels. Pairing may form and dissociate; partial matches and folded structures can compete with the desired junction.

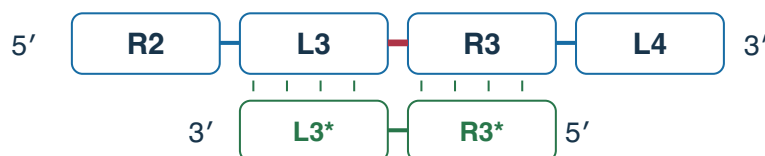
1 Separate



2 Annealed; nick remains



3 Ligated; backbone continuous



Dashed contacts = pairing. Red closure = new backbone bond. Domains not to scale.

Figure 2.4: Three states of one intended junction: separated fragments, an annealed complex with a nick, and a ligated continuous upper backbone. Dashed vertical contacts denote noncovalent pairing. Red backbone closure denotes a different chemical event. Domain widths and end-group placement are schematic.

At a properly aligned nick, T4 DNA ligase can join a 3' hydroxyl to a neighboring 5' phosphate through a phosphodiester bond.[3] The splint is an alignment device; it does not become an extra vertex code inserted into the upper strand. If the duplex later separates, the ligated upper fragments remain connected because a backbone bond has been formed.

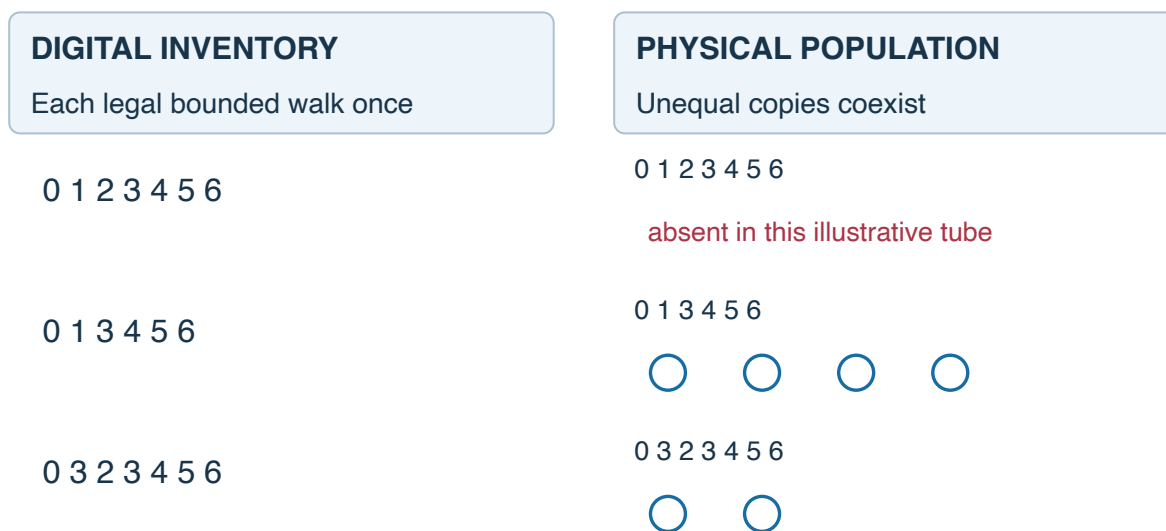
A drawing can therefore be tested mechanically. Ask which two atoms' terminal groups are being brought together, which strand carries the continuous product, and which marks mean pairing rather than covalent connection. A line joining labels without a direction or bond meaning cannot answer these questions.

Complementarity alone is not a quantitative prediction of productive binding. Neighboring base context and solution conditions affect duplex thermodynamics.[12] A ten-base domain

does not have a universal melting temperature independent of its sequence and surroundings. Chapter 6, *Hybridization thermodynamics*, develops that physics. Here the domain model establishes the intended geometry, not a validated sequence-design certificate.

2.7 A molecular population has multiplicity and missing members

The paper’s preparation used edge fragments and internal complementary vertex splints. The input inventory included 50 pmol of each species in a 100- μ L ligation mixture, with a four-hour room-temperature ligation step.[1] These historical details anchor the physical scale; they are not a modern replication protocol.



The tube is hypothetical. Selection cannot recover an absent candidate.

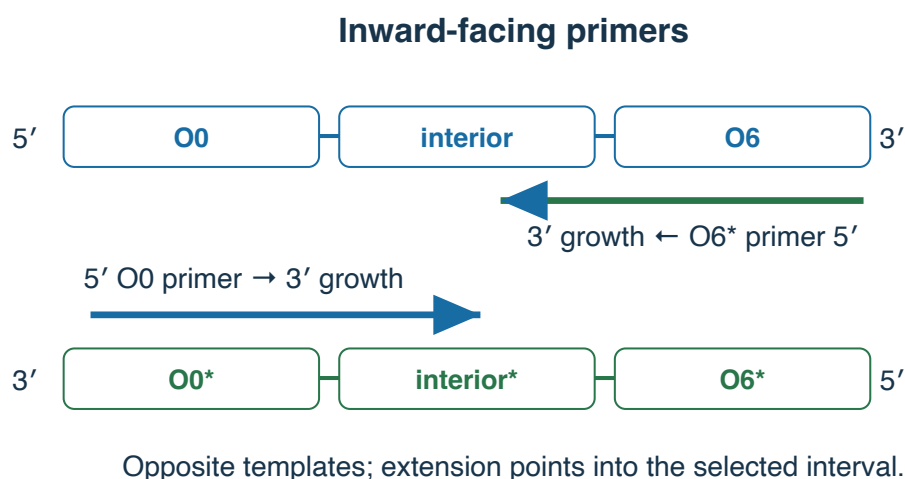
Figure 2.5: Digital enumeration lists every candidate within a declared bound; a physical preparation contains unequal multiplicities and can miss a witness entirely. The illustrated tube is deliberately hypothetical and is not a reconstruction of measured abundance.

Let $N(P)$ count copies of route P . Total material, $\sum_P N(P)$, differs from support, $\{P : N(P) > 0\}$. If all material occupies the wrong support, more copies do not solve the search problem. Equal input amounts of individual oligos do not make every assembled route equally likely: routes use different combinations of fragments and compete through physical processes.

Our program enumerates legal walks of one through eight vertices, starting anywhere. It obtains 1,583 distinct bounded walks. The real reaction is neither that exhaustive inventory nor a uniform random draw from it. In particular, cycles in the graph permit arbitrarily long abstract walks; eight is an explicit software boundary, not a chemically enforced maximum.

2.8 Endpoint PCR: orientation turns recognition into enrichment

The historical endpoint primers were associated with O_0 and the complement of O_6 . [1] To understand their logical role, follow a double-stranded candidate. The O_0 primer binds a complementary start region and extends toward the interior; the complementary- O_6 primer binds the other strand near the finish and extends inward in the opposite direction. Both new strands grow 5' to 3'.



Schematic enrichment, not exact cycle counts or guaranteed endpoint specificity.

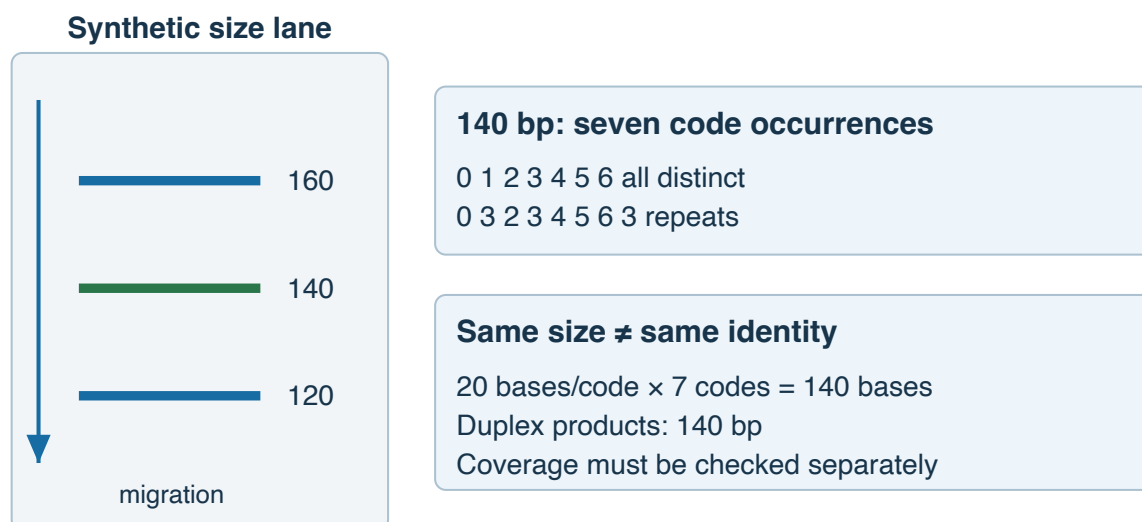
Figure 2.6: Opposed template strands and inward-facing endpoint primers. Polymerase extension grows from the primer's 3' end. Repeated separation, annealing and extension enrich a compatible product family. This is an orientation diagram, not a thermal-cycling recipe.

PCR uses repeated strand separation and primer-directed copying to amplify a selected interval.[4] Ideal doubling illustrates its power but should not be assumed for every molecule in every cycle. A template with one effective primer site, off-target binding, incomplete extension, or unequal amplification efficiency can behave differently from the ideal two-site target.

The software endpoint predicate retains an existing route unchanged or removes it. PCR can create additional copies and can select an interval within a longer template. Thus endpoint filter names the intended computational role, not a literal set-intersection operation performed without representation assumptions. A strong band means that some sequence amplified successfully; it does not independently certify a Hamiltonian order.

2.9 Size selection: seven occurrences are not seven identities

Gel electrophoresis separates charged molecules in a matrix under an electric field; size is a major determinant of migration under appropriate conditions.[5] The historical workflow recovered the 140-bp fraction, with additional purification and amplification steps. Our conceptual pipeline compresses those repeated physical operations into one ideal length predicate.



Band spacing is conceptual, not calibrated gel mobility or measured intensity.

Figure 2.7: Synthetic 120-, 140- and 160-bp bands illustrate a size fraction. The two listed seven-occurrence routes both encode 140 bases under the full-endpoint convention, despite different coverage. Positions and intensities are schematic, not experimental gel data.

A size fraction cannot distinguish 0, 1, 2, 3, 4, 5, 6 from 0, 3, 2, 3, 4, 5, 6 on length alone. Both contain seven code occurrences. The second needs a vertex-presence test to expose the missing 1. Conversely, presence alone does not exclude an overlong walk that includes every vertex plus a repetition.

This is a useful general design pattern: two weak observables can jointly imply a stronger property. Length supplies a finite budget of positions; coverage requires every identity to occupy one. Their conjunction establishes uniqueness of occurrence only when both observables are trustworthy and the representation uses the declared fixed-length codes.

2.10 Two bead operations that must not be conflated

The detailed affinity procedure has a preparatory step and a selection step. In preparation, one PCR strand was biotin-labeled through a complementary- O_6 primer and immobilized on streptavidin-coated particles. Denaturation released the other, unbiotinylated strand into solution. That free strand was the material needed for subsequent sequence recognition.

In vertex selection, the bead carries a biotinylated complementary vertex probe. A target strand containing the corresponding code can hybridize to this probe. Unbound material is washed away; denaturation releases retained target into solution for the next selection. The sequence-recognition role belongs to the probe, not to biotin.[1]



Biotin attaches. Complementarity recognizes. The retained phase changes with the step.

Figure 2.8: Preparation and selection retain different molecular relationships. First recover the unbiotinylated strand from an immobilized duplex. Then recognize a vertex through a complementary bead-bound probe, wash, and release the selected target. The diagram tracks which material remains attached and which returns to solution.

The distinction can be derived without memorizing an equipment list. Ask what is covalently attached to biotin, what is held only by hybridization, and which phase is retained after separation. During preparation, the free forward strand is wanted. During washing, a target must remain associated with the probe to avoid removal. During final release, that same association must be disrupted so the target can proceed.

2.11 Sequential presence tests and their exact logical mirror

Once endpoints are established, the internal vertices 1 through 5 can be tested in sequence. An ideal coverage filter for vertex i is

$$\mathcal{F}_i(S) = \{P \in S : i \in P\}.$$

These ideal filters commute because intersections commute. Laboratory versions need not have order-independent yields: losses, concentrations, carryover and differing recognition properties can make the physical sequence matter.

Stage	Legal inventory	With pseudo-path
generated	1583	1584
endpoints	9	10
length	2	3
contains 1	1	2
contains 2	1	2
contains 3	1	2
contains 4	1	2
contains 5	1	2
verified	1	1

Figure 2.9: Exact counts from the bounded-walk program. The second column uses only graph-valid generation. The third injects one illegal-edge permutation. It survives coverage but fails final verification. These are digital candidate counts, not measured molecule yields.

The clean inventory narrows from 1,583 walks to nine endpoint-compatible walks, then two seven-occurrence walks. Requiring vertex 1 removes the repeated-3 walk, leaving the unique witness. The later presence filters do not change this particular digital count, but their redundancy on this instance does not make coverage tests unnecessary in the general construction.

Inject the pseudo-path 0, 1, 2, 4, 3, 5, 6. The endpoint and length counts become ten and three; after testing vertex 1, two candidates remain. Both contain every required identity. Only the independent adjacency check rejects the pseudo-path. This controlled counterexample shows exactly where the soundness argument relies on generation.

2.12 One computation, three linked representations

Logical condition	Molecular property	Laboratory operation
permitted adjacency	oriented code overlap	anneal + ligate
correct endpoints	opposed primer sites	PCR enrichment
seven occurrences	140-bp product	gel fraction recovery
contains vertex i	probe-complement site	affinity selection
valid witness	resolved ordered code	decode + graph check

Figure 2.10: Aligned logical, molecular and laboratory descriptions. Read a row as a proposed implementation relation, not an equality of mathematical and physical operations. Sequence integrity, specificity, recovery and readout resolution are assumptions at those interfaces.

Let ϕ decode a well-formed molecule into a route, and extend it elementwise to a multiset decoder Φ . For a molecular population M , an ideal implementation of a multiset filter F would satisfy

$$\Phi(\text{select}(M)) = F(\Phi(M))$$

with suitable multiset conventions. A real operation is better represented by a distribution over outcomes: it can lose a desired molecule, retain an undesired one, amplify copies or produce unreadable material. The useful question is then the probability of a correct, recoverable result under a stated physical model.

This viewpoint separates **soundness** from **completeness**. Soundness asks whether an accepted decoded candidate really is a witness. Completeness asks whether an existing witness is retained and detected. Strong purification may improve the fraction of valid survivors while reducing the chance that any survivor remains.

2.13 Graduated PCR: a position projection, not sequencing

Graduated PCR used separate reactions combining the start-associated primer with a primer associated with each later vertex. Product lengths supported positions in the route.^[1] In our idealized code-length model, an occurrence at zero-based position j gives a product of $20(j + 1)$ bp from the start.



Figure 2.11: Synthetic graduated-PCR-style length projections. The pure witness gives 40, 60, 80, 100, 120 and 140 bp in lanes 1 through 6. A mixture of the witness, short route and repeated-3 route produces unions of possible lengths. This is not the historical gel photograph and makes no intensity or detection-efficiency claim.

Formally, for a population support S define

$$B_i(S) = \bigcup_{P \in S} \{20(j+1) : P_j = i\}.$$

The map $S \mapsto (B_1, \dots, B_6)$ discards molecular identity. Different multiplicities have exactly the same projection, and bands in different lanes are not tagged as belonging to the same route. Additional isolation and suitable readout may therefore be necessary before a complete order can be checked.

A positive, resolved witness verifies existence by Equation 2.1. No detectable band is a different observation: it may mean no witness, missing initial coverage, cumulative loss, or insufficient detection. Without controlling these alternatives, nothing detected cannot be promoted to a proof of nonexistence.

2.14 Executable reconstruction and independent checks

Run `python examples/deep/ch02.py`. The source exports generation, individual predicates, a staged pipeline, an independent permutation oracle, a readout projection and a retention sensitivity calculation. The code below is extracted from the tested file rather than maintained as a separate implementation.

```

1 def generate_candidates(max_vertices=8, vertices=VERTICES, edges=EDGES):
2     """Finite exhaustive walk inventory, all starts, repeats allowed."""
3     if type(max_vertices) is not int or max_vertices < 1:
4         raise ValueError("positive integer bound required")
5     if not vertices or len(set(vertices)) != len(vertices):
6         raise ValueError("distinct nonempty vertices required")
7     if any(u not in vertices or v not in vertices for u, v in edges):
8         raise ValueError("edge outside vertex set")
9     frontier = [(v,) for v in vertices]
10    result = []
11    for _ in range(max_vertices):
12        result.extend(frontier)
13        frontier = [p + (v,) for p in frontier for v in vertices if (p[-1], v) in edges]
14    return result

```

The generator deliberately extends a frontier of legal walks, not permutations. Repetitions must remain possible so later filters have work to do. At successive lengths 1 through 8, the inventory contains

7, 14, 25, 49, 99, 199, 397, 793

walks. Their sum is 1,583. Counting paths by length and counting survivors by predicate test different aspects of the implementation.

```

1 def pipeline(pool):
2     stages = [("generated", list(pool))]
3     stages.append(("endpoints", filter_start_end(stages[-1][1])))
4     stages.append(("length", filter_length(stages[-1][1])))
5     for vertex, kept in filter_all_vertices(stages[-1][1], range(1, 6)):
6         stages.append((f"contains {vertex}", kept))
7     stages.append(("verified", [p for p in stages[-1][1] if is_witness(p)]))
8     return stages

```

The oracle starts from a different representation: all permutations of the five internal vertices. Agreement between two implementations is more informative when they do not share the same candidate generator. Tests additionally remove each witness edge, inject pseudo-path copies, validate orientation arithmetic, and check that multiplicity is preserved by the selection functions.

This remains a logical mirror. It has no rate constants, thermodynamic ensemble, PCR efficiency, bead binding kinetics or calibrated gel resolution. Adding those would require explicit models and experimentally supported parameters, not merely more Python loops.

2.15 Purity can improve while recovery becomes fragile

Suppose there are initially W_0 true copies and F_0 false candidates. In a deliberately simple model, each surviving true copy passes each next selection with conditional probability q , and each false candidate with probability f . After k stages,

$$\mathbb{E}[W_k] = W_0 q^k, \quad \mathbb{E}[F_k] = F_0 f^k.$$

These expectations do not require independence across molecules. To derive the probability of losing every true copy, additionally assume independent survival across initial copies:

$$\Pr(W_k = 0) = (1 - q^k)^{W_0}. \quad (2.2)$$

Selections	Expected true copies	Expected false copies
0	3	1000
1	2.4	100
2	1.92	10
3	1.536	1
4	1.2288	0.1
5	0.98304	0.01

Three initial true copies; $q=0.8$. After five stages: $P(\text{no true copy}) \approx 0.304$.

Figure 2.12: Original sensitivity model with $W_0 = 3$, $F_0 = 1000$, $q = 0.8$ and $f = 0.1$. Expected false material falls rapidly, but the probability of no true survivor after five stages is about 0.304 under independent-copy survival. Fractional expectations are not fractional physical molecules.

After five stages, expected true copies are 0.98304 and expected false copies 0.01. A small expected false count is not a guarantee that the sample contains a witness. Nor is the ratio of expectations generally equal to the expected purity ratio: $\mathbb{E}[W]/\mathbb{E}[W + F]$ is not usually $\mathbb{E}[W/(W + F)]$, especially when empty outcomes are possible.

Real molecules can share environmental failures, violating independent-copy survival. A failed transfer can lose many copies together. The toy calculation identifies sensitivity to cumulative retention; it does not estimate Adleman's false-negative probability.

2.16 Material, time and the conditional factorial argument

Use an engineering resource vector

$$R = (N_{\text{molecules}}, V_{\text{solution}}, T_{\text{lab}}, N_{\text{operations}}, P_{\text{error}}, C_{\text{readout}}).$$

It is an accounting framework, not a universal complexity definition. The components have different units and cannot be collapsed into speed without a chosen application and cost model.

Historical input inventory50 pmol/species → about 3.01×10^{13} oligos/species

14 edge species + 5 internal splints; not 19 completed routes

Vertices n	Fixed-endpoint orders	Assumption
7	120	complete directed graph
10	40,320	complete directed graph
15	6,227,020,800	complete directed graph
20	6,402,373,705,728,000	complete directed graph

Search-space count $\neq$ sampled route support $\neq$ measured reagent concentration.

Figure 2.13: Input molecules and candidate orders belong to different accounts. The historical reagent amount converts to about 3.01×10^{13} oligos per species. The factorial column counts interior permutations only for a complete directed graph with fixed endpoints; it is not an observed route distribution or a universal DNA lower bound.

Using the exact Avogadro constant,[7]

$$50 \times 10^{-12} \text{ mol} \times 6.02214076 \times 10^{23} \text{ mol}^{-1} = 3.01107038 \times 10^{13}.$$

There are fourteen edge species and five internal splint species in this inventory. These are reagent molecules; several edge fragments contribute to one complete route, and splints play an alignment role. Treating every oligo as an independently encoded full candidate would overstate coverage.

With fixed endpoints in a complete directed graph, the interior order space has $(n - 2)!$ members. Under independent uniform sampling of those orders with one witness, its miss probability after M complete-route samples is $(1 - 1/(n - 2)!)^M$. In the more realistic nonuniform abstraction, replace the reciprocal by the total probability p of generating a witness:

$$M \geq \frac{\ln \delta}{\ln(1 - p)}$$

for miss probability at most δ , with $0 < p, \delta < 1$. Neither formula tells us p for the historical chemistry. Sparse edges, unequal assembly probabilities and alternative algorithms change the account.

The reported laboratory workflow took about seven days, not merely the time of one parallel chemical reaction.[1] Preparation, transfers, amplification, recovery and readout are part of elapsed work. The demonstration does not establish asymptotically efficient NP-complete search, fault-free computation, or automatic superiority over silicon.

2.17 What the experiment opened, and what changed later

The bounded achievement is already substantial: molecular representations and selective operations participated in solving a small combinatorial instance. This links an abstract

problem to physical information processing. It does not require an unsupported general claim about competitive runtime.

Tradition	State / operation	Question opened
SAT filtering	assignments / selection	which clauses hold?
Formal DNA systems	strings / formal operators	what can be computed?
Molecular automata	state + input / transitions	can reactions follow rules?
Self-assembly	tiles / local attachment	can shape encode a program?
Strand displacement	signals / strand exchange	can chemistry compose gates?

Figure 2.14: A conceptual map of later research traditions, organized by state and operation. This is neither a causal genealogy nor a ranking of superiority. The mechanisms differ; subsequent chapters must derive their own correctness and resource arguments.

Lipton explored SAT-style molecular search as a proposal,[13] while formal work studied computational power under specified DNA operations.[14] Molecular automata used biochemical transitions;[8] self-assembly organized local interactions into structures;[11] strand-displacement circuits made recognition and exchange into composable signal operations.[15] These are not all variants of filtering a pre-existing factorial route pool.

Chapter 3, *Combinatorial search and complexity*, next separates instance difficulty, verification cost, reductions and physical resource accounting. The later thermodynamic and kinetic chapters then examine whether the symbolic operations assumed here are physically reliable. The separation is intentional: a correct combinatorial proof and a workable molecular implementation must both be earned.

2.18 Exercises

1. Prove uniqueness of the historical witness without enumerating 120 permutations. Identify the graph feature that forces the final block.
2. Remove edge $3 \rightarrow 4$. What must happen to the independent oracle? Why cannot more copies repair this instance?
3. Derive the lengths of the direct $0 \rightarrow 6$ fragment, a three-vertex route, and a seven-vertex route. Repeat with incomplete terminal half-codes.
4. Calculate O_{23} and the aligned complement of O_3 . Explain why reversing the latter changes its direction convention.
5. A tube retains all seven identities in a 160-base route. Give the logical reason it is not yet a Hamiltonian certificate.
6. Explain which material is retained in strand preparation, washing during vertex capture, and target release. Identify the recognition element.

7. Add ten copies of the illegal permutation to the Python inventory. Predict and then check all stage counts.
8. Derive Equation 2.2; state separately the assumptions needed for expected count and zero-survivor probability.
9. Show that the graduated-band projection cannot recover route multiplicity. What further ambiguity arises when several routes contribute different lane positions?
10. For fixed endpoints and $n = 10$ in a complete directed graph, count interior orders. Under one uniformly sampled witness, derive the number of independent complete-route draws needed for a 99 percent hit probability.
11. Construct a physical false positive and a false negative at different stages. Explain why independently verifying one decoded witness addresses only one of them.
12. Propose an extension of the digital mirror with unequal generation probabilities. Specify what can be computed exactly and what would still require physical evidence.

2.19 Worked solutions and research boundaries

1. Only 3 enters 4 and only 4 enters 5. Finishing requires 6 last. Leaving 5 for 1 or 2 after 3 and 4 have been used cannot reach 6 without revisiting the final block. Thus the suffix is 3,4,5,6. The remaining vertices force the prefix 0,1,2.
2. The oracle becomes empty because 4 loses its only incoming edge. Additional material duplicates candidates but cannot create a graph-valid missing transition. A chemical product crossing that transition would be a representation error, not a solution.
3. Complete endpoints give 40, 60 and 140 bases respectively: each visited vertex contributes a complete twenty-base code. If only an interior-facing half of each terminal code remains, subtract twenty bases in total, giving 20, 40 and 120. The corresponding duplex lengths use bp.
4. Take the last ten bases of O_2 followed by the first ten of O_3 : GTATATCCGAGCTATTCGAG. Basewise complementing O_3 gives CGATAAGCTCGAATTTTCGAT when printed aligned 3' to 5'. Reversal is additionally necessary to list that partner 5' to 3'.
5. At twenty bases per occurrence, 160 bases represent eight occurrences. Containing seven identities then requires at least one repetition, assuming only those identities occur. Coverage without the correct occurrence count does not imply distinctness.
6. Preparation keeps the unbiotinylated free strand after denaturation. Washing keeps targets still hybridized to the bead-bound probe. Release keeps those selected targets in solution after disrupting that pairing. Complementary sequence provides recognition; biotin–streptavidin provides attachment.
7. Multiplicity matters. Counts become 1,593; 19; 12; then 11 after each of the five presence tests; and finally 1 after verification. Ten identical pseudo-paths should remain ten copies until a predicate rejects them. Converting the pool to a set would conceal the test.
8. A given true copy survives all stages with probability q^k under constant conditional retention. Summing its survival indicator over W_0 copies gives the expectation without requiring independent copies. Independence across copies gives the product of their failure probabilities, $(1 - q^k)^{W_0}$.

9. Set union is idempotent: adding another copy of a route adds no new band positions. In a mixture, a band in lane 2 and a band in lane 3 can originate from different molecules. The projection does not retain which positions co-occur on one route.
10. There are $8! = 40,320$ interior orders. The exact minimum is the ceiling of $\ln(0.01)/\ln(1 - 1/40320)$. Use the logarithmic expression rather than subtracting an approximate expectation from one. It describes independent uniform complete-route samples, not starting oligos.
11. Nonspecific retention of a pseudo-path can give a false positive; losing the only witness during a transfer can give a false negative. A decoded witness checked against the graph rejects the former. It cannot establish that absent material was never generated or that a negative instance has been proved.
12. Assign probabilities to outgoing transitions and an explicit stopping rule, then propagate probability mass over bounded walks. Exact dynamic programming can compute the resulting distribution and witness mass for that model. Transition probabilities, stopping behavior and their transferability to chemistry still need measurement; a convenient probability table is not experimental validation.

Chapter 1 glossary

Walk A vertex sequence respecting directed edges; repeats are allowed. See Section 1.3.

Hamiltonian path A simple path containing every graph vertex exactly once. See Section 1.3.

Decision problem A problem whose required answer is yes or no. See Section 1.3.

Domain A contiguous sequence assigned a recognition role in a molecular design. See Section 1.4.

Reverse complement The complementary DNA sequence rewritten in the same 5-prime to 3-prime convention. See Section 1.4.

Multiset A collection retaining how many copies of each object occur. See Section 1.5.

Support The distinct objects present with nonzero multiplicity. See Section 1.5.

Hybridization Association through complementary sequence interactions. See Section 1.7.

Ligation Covalent connection between suitable adjacent backbone ends. See Section 1.7.

PCR Primer-directed cyclic DNA amplification. See Section 1.7.

Affinity selection Retention supported by a probe recognizing a target subsequence. See Section 1.7.

Certificate A proposed witness whose correctness can be checked against the instance. See Section 1.8.

Sampling failure Missing a target event in the declared number of draws. See Section 1.9.

Chapter 2 glossary

Vertex splint Complementary strand that aligns adjacent edge fragments at a shared vertex; it does not insert an additional code into the product backbone.

Nick Discontinuity in one strand's backbone within an otherwise paired structure. Pairing across the junction does not itself seal the nick.

Affinity selection Recognition and retention using a binding interaction. Here a complementary probe recognizes a vertex code; biotin attaches the probe to a recoverable support.

Graduated PCR A set of primer-dependent amplifications whose product lengths can support position inference. The lane projection does not retain joint identity across a mixture.

Soundness and completeness Correctness of accepted witnesses versus recovery and detection of existing witnesses. A physical implementation must consider both.

Support and multiplicity Which route types are present versus how many copies of each exist. Neither is determined solely by total reagent molecules.

Bibliography

- [1] Leonard M. Adleman. Molecular computation of solutions to combinatorial problems. *Science* 266 (1994), 1021–1024. [Source link](#).
- [2] Bruce Alberts et al. The Structure and Function of DNA. *Molecular Biology of the Cell*, 4th edition (2002). [Source link](#).
- [3] New England Biolabs. T4 DNA Ligase, product description. [Source link](#).
- [4] National Human Genome Research Institute. Polymerase Chain Reaction Fact Sheet. [Source link](#).
- [5] National Human Genome Research Institute. Electrophoresis. [Source link](#).
- [6] Frank Pfenning. Lecture 36: The Hamiltonian Path Problem. Carnegie Mellon University, 15-453. [Source link](#).
- [7] Bureau International des Poids et Mesures. SI base unit: mole. [Source link](#).
- [8] Yaakov Benenson et al. Programmable and autonomous computing machine made of biomolecules. *Nature* 414 (2001), 430–434. [Source link](#).
- [9] David Soloveichik, Georg Seelig, and Erik Winfree. DNA as a universal substrate for chemical kinetics. *PNAS* 107 (2010), 5393–5398. [Source link](#).
- [10] Lulu Qian and Erik Winfree. Scaling up digital circuit computation with DNA strand displacement cascades. *Science* 332 (2011), 1196–1201. [Source link](#).
- [11] Erik Winfree, Furong Liu, Lisa A. Wenzler, and Nadrian C. Seeman. Design and self-assembly of two-dimensional DNA crystals. *Nature* 394 (1998), 539–544. [Source link](#).
- [12] John SantaLucia Jr. A unified view of polymer, dumbbell, and oligonucleotide DNA nearest-neighbor thermodynamics. *PNAS* 95 (1998), 1460–1465. [Primary article](#).
- [13] Richard J. Lipton. DNA solution of hard computational problems. *Science* 268 (1995), 542–545. [Primary abstract](#).
- [14] Dan Boneh, Christopher Dunworth, Richard J. Lipton and Jiri Sgall. On the computational power of DNA. *Discrete Applied Mathematics* 71 (1996), 79–94. [Author publication record](#).
- [15] Georg Seelig, David Soloveichik, David Y. Zhang and Erik Winfree. Enzyme-free nucleic acid logic circuits. *Science* 314 (2006), 1585–1588. [Author manuscript record](#).

Index

affinity selection, 22

biotin, 22

completeness, 25

decision problem, 4

Gel electrophoresis, 8

graduated PCR, 25

Hamiltonian path, 3

Hybridization, 7

Ligation, 7

multiset, 5

Polymerase chain reaction, 8

readout ambiguity, 25

reverse complement, 5

soundness, 25

vertex splint, 22

walk, 3